Why three? Trees of rationals from $x \mapsto x+1$ and an involution

Arjun Pemmasani
[Placeholder for the author: affiliation and email address]

Abstract

Start at 0 and apply the maps $x \mapsto x+1$ and $x \mapsto -1/x$. Problem 137 of Kagey's Open Problem Collection asks why the number $a(n)$ of rationals first reached after n steps satisfies $a(n) = a(n-1) + a(n-3)$. We show that the 3 is the order of the map $x \mapsto -1/(x+1)$. To see this we replace $-1/x$ by $g_m(x) = -1/(mx)$. For $m = 2$ and $m = 3$ the composite $g_m(x+1)$ has order $q = 4$ and $q = 6$, every rational is still reached, and the counts satisfy $a(n) = a(n-1) + a(n-3) + \cdots + a(n-2q+3)$ for $n \geq 2q-2$. More generally, for every $q \geq 3$ the tree of $X \mapsto X + 2\cos(\pi/q)$ and $X \mapsto -1/X$ has rank generating function $(1 + t^2 + \cdots + t^{2q-4} - t^{2q-3})/(1 - t - t^3 - \cdots - t^{2q-3})$. The proof gives the rank of each point from a continued fraction whose digits obey one rule. No $q-2$ digits in a row after the first may equal 1, with one exception at the root. For $m \geq 4$ the composite has infinite order, the counts are Fibonacci numbers, and some rationals are never reached.

2020 Mathematics Subject Classification: 11B37, 05A15, 11A55, 20H10.
Keywords: tree of rationals, linear recurrence, Hecke group, Rosen continued fraction, generating function.

1 Introduction

1.1 The question

Let $f(x) = x+1$ and $g(x) = -1/x$. Start at 0 and apply f and g in any order, never applying g to 0. Every rational is reached, and the *rank* of a rational is the least number of moves needed to reach it. Kimberling sorted the rationals by rank in OEIS A226247 [11], and Kagey asked whether the number $a(n)$ of rationals of rank n satisfies $a(n) = a(n-1) + a(n-3)$ [6, 5]. In earlier work [14] we proved this for $n \geq 4$, together with a formula for the rank and a description of the tree. Kimberling and Moses had stated the same recurrence for the version of the tree with root 1 [7, Example 4.2].

This paper asks where the 3 comes from. The answer is short. The map

$$g(f(x)) = -\frac{1}{x+1}$$

has order 3, since $x \mapsto -1/(x+1) \mapsto -(x+1)/x \mapsto x$. To test this reading we change g so that $g \circ f$ has a different order and see what happens to the recurrence. Keep $f(x) = x+1$ and the root 0, and for an integer $m \geq 1$ put

$$g_m(x) = -\frac{1}{mx}.$$

Problem 137 is the case $m = 1$. The composite $g_m(f(x)) = -1/(m(x+1))$ is a Möbius map with trace m and determinant m. It has order 3, 4 and 6 for $m = 1$, 2 and 3, and infinite order for $m \geq 4$ (Lemma 3.2). We write q for this order. Table 1 summarizes what we prove about each case.

m	q	points of the tree	generating function of $a(n)$	recurrence
1	3	all of $\mathbb{Q}$	$\dfrac{1+t^2-t^3}{1-t-t^3}$	$a(n)=a(n-1)+a(n-3)$, $n\ge 4$
2	4	all of $\mathbb{Q}$	$\dfrac{1+t^2+t^4-t^5}{1-t-t^3-t^5}$	$a(n)=a(n-1)+a(n-3)+a(n-5)$, $n\ge 6$
3	6	all of $\mathbb{Q}$	$\dfrac{1+t^2+t^4+t^6+t^8-t^9}{1-t-t^3-t^5-t^7-t^9}$	$a(n)=\sum_{j=0}^{4} a(n-2j-1)$, $n\ge 10$
≥ 4	∞	not all of $\mathbb{Q}$	$\dfrac{1}{1-t-t^2}$	$a(n)=a(n-1)+a(n-2)$, $n\ge 2$

Table 1: The trees of $f(x)=x+1$ and $g_m(x)=-1/(mx)$ with root 0. The row $m=1$ is [14]. In the third row the sum is $a(n-1)+a(n-3)+a(n-5)+a(n-7)+a(n-9)$. For $m\le 3$ each recurrence fails at the first n below the stated range.

So the 3 in Problem 137 is the order of $g\circ f$, and the recurrence for order q has the $q-1$ lags $1,3,5,\dots,2q-3$. The family also shows why the more obvious generalization $x\mapsto x+k$ with $x\mapsto -1/x$ is not interesting. Putting $x=ky$ turns it into the tree of $y\mapsto y+1$ and $y\mapsto -1/(k^2y)$, which is the case $m=k^2\ge 4$, and its counts are Fibonacci numbers.

The cases $m=2$ and $m=3$ give new trees of rationals. They are the only other values of m for which the tree is all of $\mathbb{Q}$ (Theorem D and Theorem E). Their counts start

$$(m=2)\qquad 1,1,2,3,5,7,11,18,28,44,69,108,170,267,419,658,1033,\dots,$$
$$(m=3)\qquad 1,1,2,3,5,8,13,21,34,54,87,141,227,366,590,951,1533,\dots,$$

and neither sequence is in the OEIS. Figure 1 shows the first ranks of the trees for $m=1$ and $m=2$.

1.2 The trees for every q

The proofs do not use m at all. They use only the order q. With $X=\sqrt{m}\,x$ the maps become $X\mapsto X+\sqrt{m}$ and $X\mapsto -1/X$, and for $m=1,2,3$ we have $\sqrt{m}=2\cos(\pi/q)$. So for every $q\ge 3$ we put

$$\lambda=\lambda_q=2\cos\frac{\pi}{q},\qquad F(X)=X+\lambda,\qquad G(X)=-\frac{1}{X},$$

and study the tree of F and G with root 0, which we call the *λ_q-tree*. For $q=3,4,6$ it is the tree of f and g_m with $m=1,2,3$, scaled by $\sqrt{m}$ (Lemma 2.3). For $q=5$ it lives in $\mathbb{Q}(\sqrt{5})$, and for most larger q in a number field of higher degree. The maps F and G generate Hecke's group H_q [2], and $G\circ F$ has order q.

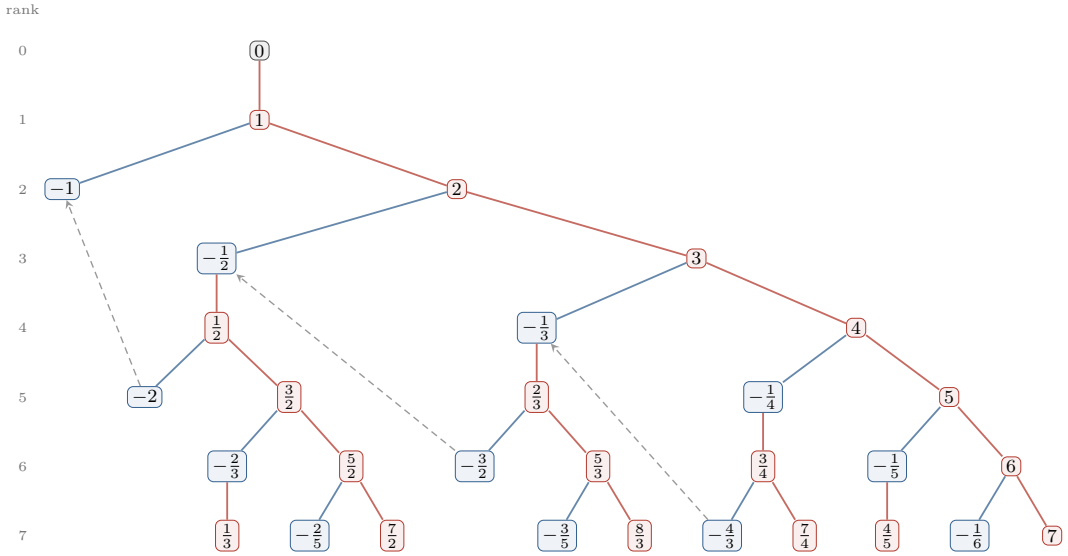
We write $d(X)$ for the rank of a point X of the λ_q-tree, $a_q(n)$ for the number of points of rank n, and $r_q(n)$ for the number of positive ones. We also put

$$D_q(t)=1-t-t^3-t^5-\cdots-t^{2q-3},\qquad N_q(t)=1+t^2+t^4+\cdots+t^{2q-4}-t^{2q-3}.$$

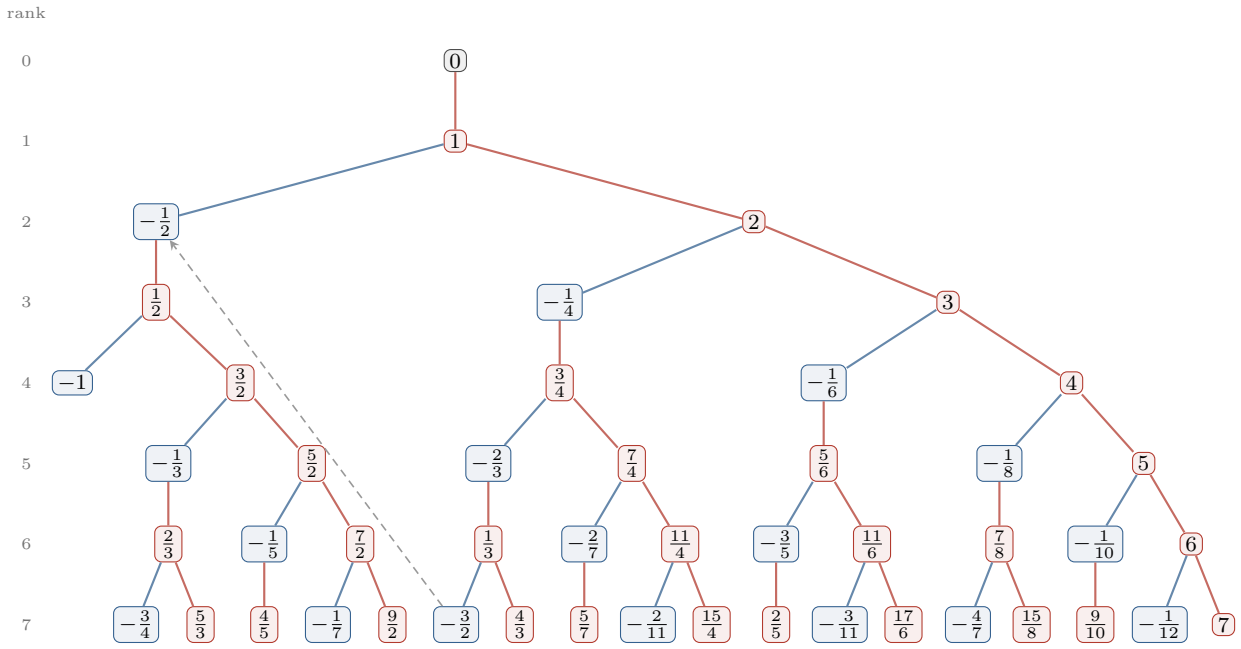
Theorems A_q, B_q and C_q below hold for every $q\ge 3$, and at $q=3$ they give the corresponding parts of Theorems A, B(ii) and C of [14].

1.3 Results

A string of integers $(c_1,\dots,c_k)$ with $k\ge 1$, $c_1\ge 0$ and $c_2,\dots,c_k\ge 1$ is *reduced* if no $q-2$ consecutive digits among $c_2,\dots,c_k$ equal 1, with one exception. If $c_1=0$, the run of 1's that starts at c_2 may



(a) $m = 1$, the tree of Problem 137.



(b) $m = 2$.

Figure 1: Ranks 0 to 7 of the trees of $f(x) = x + 1$ (red edges) and $g_m(x) = -1/(mx)$ (blue edges) for $m = 1$ and $m = 2$. Positive vertices are entered by f and negative ones by g_m (Theorem C_q). The vertices $x \leq -1$ are leaves. A dashed arrow shows f applied to a leaf $x < -1$. The value $x + 1$ is not new and its rank is lower by $2q - 3$, which is 3 for $m = 1$ and 5 for $m = 2$. The numbers on the left are ranks.

have length exactly $q-2$. Its *word* is $f^{c_1}gf^{c_2}g\cdots gf^{c_k}$, applied to 0 from right to left, and its *weight* is $c_1+\cdots+c_k+k-1$, the length of the word. For $q=3$ a reduced string has $c_2,\ldots,c_k\geq 2$, except that $c_2=1$ is allowed when $c_1=0$. This is the rule of [14, Theorem B].

Theorem B_q (the rank formula, Theorem 5.3). Let $q\geq 3$. Every point of the λ_q-tree is reached by the word of exactly one reduced string, and its rank is the weight of that string. This word is the only shortest word for the point. In coordinates, the string $(c_1,\ldots,c_k)$ gives

$$X = c_1\lambda - \cfrac{1}{c_2\lambda - \cfrac{1}{\ddots - \cfrac{1}{c_k\lambda}}},$$

read as $-1/(\cdots)$ when $c_1=0$. For $m=2,3$ every rational x has exactly one reduced string, $x=c_1+g_m\big(c_2+g_m(\cdots+g_m(c_k))\big)$, and

$$d(x) = c_1+c_2+\cdots+c_k+k-1.$$

For example, take $m=2$ and $x=\frac{3}{5}$. Then

$$\tfrac{3}{5} = 1 - \cfrac{1}{4 - \cfrac{1}{1 - \cfrac{1}{4 - \cfrac{1}{1}}}},$$

which is $1+g_2(2+g_2(1+g_2(2+g_2(1))))$. Its string is $(1,2,1,2,1)$, which is reduced for $q=4$ because no two consecutive digits after the first equal 1. So $d(\frac{3}{5})=7+4=11$.

Theorem A_q (the counts, Theorem 6.1). Let $q\geq 3$. Then

$$\sum_{n\geq 0} a_q(n)\,t^n = \frac{N_q(t)}{D_q(t)},$$

and this fraction is in lowest terms. Hence

$$a_q(n) = a_q(n-1)+a_q(n-3)+a_q(n-5)+\cdots+a_q(n-2q+3) \qquad \text{for all } n\geq 2q-2,$$

and this fails at $n=2q-3$, where the left side minus the right side is -1. The positive points have $\sum_n r_q(n)t^n = t(1+t^2+\cdots+t^{2q-6})/D_q(t)$, and $a_q(n)=r_q(n)+r_q(n-1)$ for $n\geq 1$.

The denominator depends only on q, and so does the whole count. For $m=2$ the positive counts are $r(n)=\text{A060961}(n-1)+\text{A060961}(n-3)$, where A060961 counts compositions into parts 1, 3 and 5 [10].

Theorem C_q (the tree, Corollary 3.7). Let $q\geq 3$ and define the parent map $P(X)=X-\lambda$ for $X>0$ and $P(X)=-1/X$ for $X<0$. For every point $X\neq 0$ of the λ_q-tree, $P(X)$ is the only point of rank $d(X)-1$ from which one move reaches X, and iterating P reaches 0 in exactly $d(X)$ steps. The last move of every shortest word for X is F if $X>0$ and G if $X<0$. So in Kagey's coloring, where a vertex is blue when it is reached last by G, a vertex is blue if and only if it is negative. A vertex $X>0$ has the two children $X+\lambda$ and $-1/X$, a vertex in $(-\lambda,0)$ has the one child $X+\lambda$, and a vertex $X\leq -\lambda$ is a leaf. For $X<-\lambda$ the value $X+\lambda$ appeared exactly $2q-3$ ranks earlier.

In x-coordinates the leaves are the $x \le -1$, and for $x < -1$ the value $x+1$ appeared five ranks earlier when $m = 2$ and nine ranks earlier when $m = 3$. This jump of $2q-3$ ranks is where the lag $2q-3$ of the recurrence comes from.

Theorem D (reachability, Theorem 4.1). For $m = 1, 2, 3$ every rational number is in the tree of $f(x) = x+1$ and $g_m(x) = -1/(mx)$.

For $m = 1$ this is [14, Theorem 3.6]. The group generated by f and g_m is known to move 0 to every rational for $m = 2, 3$ (Remark 3.8). Theorem D is stronger, because the tree uses only the forward moves f and g_m and never f^{-1}.

Theorem E (the free case, Theorem 7.1). Let $m \ge 4$. The tree of $f(x) = x+1$ and $g_m(x) = -1/(mx)$ has $a(n) = F_{n+1}$ points of rank n, where F_n is the Fibonacci sequence with $F_1 = F_2 = 1$. Every finite string $(c_1, \dots, c_k)$ with $c_1 \ge 0$ and $c_2, \dots, c_k \ge 1$ (a free string) gives a point, distinct strings give distinct points, and the rank of the point is the weight of the string. A vertex is blue if and only if it is negative, and no rational in $(-\infty, -2/m]$ is reached. In particular -1 and $\frac{1}{2}$ are never reached. This covers the tree of $x \mapsto x+k$ and $x \mapsto -1/x$ for every $k \ge 2$.

The growth rate of $a_q(n)$ is the largest root ψ_q of $\psi^{2q-3} = \psi^{2q-4} + \psi^{2q-6} + \cdots + 1$. These numbers increase with q and tend to the golden ratio (Proposition 6.2).

1.4 How the proofs go

The proofs follow [14] step by step. Section 3 proves that the parent map traces a shortest path, which gives Theorem C_q. The key step is the analogue of the identity $d(x-1) = d(x) + 3$ for $x < 0$, which becomes $d(X - \lambda) = d(X) + 2q - 3$. Its proof follows a path of $2q-2$ steps through the orbit of ∞ under $F \circ G$, which has length q. Section 4 proves Theorem D with a height that drops when we shift x to the nearest integer and apply g_m. Section 5 proves Theorem B_q and explains the run rule. Section 6 counts reduced strings and proves Theorem A_q, and Section 7 treats $m \ge 4$. Section 8 lists the computer checks, and Section 9 discusses related work and open questions. None of the proofs depends on the computations or on facts about Hecke groups.

2 Setting and notation

Fix $q \ge 3$ and put $\theta = \pi/q$ and $\lambda = \lambda_q = 2\cos\theta$, so that $1 \le \lambda < 2$. Let

$$F(X) = X + \lambda, \qquad G(X) = -\frac{1}{X} \quad (X \neq 0).$$

Definition 2.1. Let Γ_q be the directed graph on $\mathbb{R}$ with an *F-edge* $X \to X + \lambda$ for every X and a *G-edge* $X \to -1/X$ for every $X \neq 0$. Let $\mathcal{O} = \mathcal{O}_q$ be the set of points that can be reached from 0 along a directed path, and for $X \in \mathcal{O}$ let the *rank* $d(X)$ be the length of a shortest such path. We call $\mathcal{O}$ with the ranks the *λ_q-tree*, and put

$$a_q(n) = \big|\{X \in \mathcal{O} \mid d(X) = n\}\big|, \qquad r_q(n) = \big|\{X \in \mathcal{O} \mid X > 0,\ d(X) = n\}\big|.$$

We write a path from 0 as a composition of maps, applied from right to left. So $GF(0) = -1/\lambda$ means that we first apply F and then G. (In [14] words were read from left to right in the order the moves are made, so the words there are the reverses of the ones here.) A word may never apply G at 0. Following Kagey, a vertex $X \neq 0$ is *red* if the last move of a shortest path to it is F and *blue* if it is G. Corollary 3.7 shows that this is well defined.

Definition 2.2. For an integer $m \geq 1$, the *m-tree* is defined in the same way on $\mathbb{Q}$, with the maps $f(x) = x + 1$ and $g_m(x) = -1/(mx)$ and root 0. Its rank function is also written d, and $a(n)$ is the number of its points of rank n.

Lemma 2.3. *Let (m, q) be $(1, 3)$, $(2, 4)$ or $(3, 6)$. Then $x \mapsto \sqrt{m}\,x$ is an isomorphism from the graph of the m-tree, extended to $\mathbb{R}$, onto Γ_q. It takes f-edges to F-edges and g_m-edges to G-edges. In particular the m-tree and the λ_q-tree have the same counts, the same ranks at corresponding points, and the same colors.*

Proof. Note that $\lambda_3 = 1$, $\lambda_4 = \sqrt{2}$ and $\lambda_6 = \sqrt{3}$, so $\lambda_q = \sqrt{m}$. For $X = \sqrt{m}\,x$ we have $\sqrt{m}\,(x + 1) = X + \lambda_q$ and $\sqrt{m}\,g_m(x) = -\sqrt{m}/(mx) = -1/X$. □

So the case $q = 3$ is the tree of [14], and everything proved below for the λ_q-tree holds for the m-tree when $m = 1, 2, 3$, with λ replaced by 1. Table 2 works out the first ranks of the 2-tree by hand, in the same way as [14, Table 1].

n	moves from the rationals of rank $n-1$	rank n	$a(n)$
0	(start)	0	1
1	$0 \mapsto 1$ ($g_2(0)$ is not defined)	1	1
2	$1 \mapsto 2,\ -\frac{1}{2}$	$2,\ -\frac{1}{2}$	2
3	$2 \mapsto 3,\ -\frac{1}{4};\quad -\frac{1}{2} \mapsto \frac{1}{2},\ 1_{[1]}$	$3,\ -\frac{1}{4},\ \frac{1}{2}$	3
4	$3 \mapsto 4,\ -\frac{1}{6};\quad -\frac{1}{4} \mapsto \frac{3}{4},\ 2_{[2]};\quad \frac{1}{2} \mapsto \frac{3}{2},\ -1$	$4,\ -\frac{1}{6},\ \frac{3}{4},\ \frac{3}{2},\ -1$	5
5	$4 \mapsto 5,\ -\frac{1}{8};\quad -\frac{1}{6} \mapsto \frac{5}{6},\ 3_{[3]};\quad \frac{3}{4} \mapsto \frac{7}{4},\ -\frac{2}{3};$ $\frac{3}{2} \mapsto \frac{5}{2},\ -\frac{1}{3};\quad -1 \mapsto 0_{[0]},\ \frac{1}{2}_{[3]}$	$5,\ -\frac{1}{8},\ \frac{5}{6},\ \frac{7}{4},\ -\frac{2}{3},\ \frac{5}{2},\ -\frac{1}{3}$	7

Table 2: Ranks 0 to 5 of the 2-tree. In row n, "$x \mapsto u,\ v$" means $f(x) = u$ and $g_2(x) = v$ for a rational x of rank $n - 1$. Gray values were found earlier, and the subscript in brackets is their rank.

Already here the patterns of [14] show up with new numbers. Positive values are reached by f and negative ones by g_2. The leaf -1 appears at rank 4, and $f(-1) = 0$ goes back to the root. In general -1 has rank $2q - 4$ (Proposition 3.5(c)), which is 2 for $m = 1$ and 4 for $m = 2$.

3 The parent map and the distance formula

3.1 The orbit of infinity

Let $\tau = F \circ G$, so that $\tau(Y) = \lambda - 1/Y$, with $\tau(0) = \infty$ and $\tau(\infty) = \lambda$. Put

$$\beta_i = \tau^{\,i}(\infty) \quad (i \geq 0), \qquad U_i = \frac{\sin((i+1)\theta)}{\sin\theta} \quad (i \geq -1).$$

Lemma 3.1. *(a) $\beta_i = \sin((i+1)\theta)/\sin(i\theta)$ for $1 \leq i \leq q-1$.*

(b) $\infty = \beta_0 > \beta_1 = \lambda > \beta_2 > \cdots > \beta_{q-2} = 1/\lambda > \beta_{q-1} = 0$.

(c) τ^q is the identity, and $\tau(1/\lambda) = 0$.

(d) For $0 \leq i \leq q-3$, τ maps $(\beta_{i+1}, \beta_i]$ increasingly onto $(\beta_{i+2}, \beta_{i+1}]$.

Proof. We have $U_{-1} = 0$, $U_0 = 1$, $U_1 = \lambda$ and $U_{i+1} = \lambda U_i - U_{i-1}$, by the identity $\sin((i+2)\theta) + \sin(i\theta) = 2\cos\theta\,\sin((i+1)\theta)$. The matrix of τ is $\left(\begin{smallmatrix}\lambda & -1\\ 1 & 0\end{smallmatrix}\right)$, and by induction the matrix of τ^i is

$$\begin{pmatrix} U_i & -U_{i-1} \\ U_{i-1} & -U_{i-2} \end{pmatrix} \qquad (i \geq 1).$$

Indeed, multiplying this by the matrix of τ on the right gives first column $(\lambda U_i - U_{i-1},\ \lambda U_{i-1} - U_{i-2}) = (U_{i+1}, U_i)$ and second column $(-U_i, -U_{i-1})$. Hence $\beta_i = U_i/U_{i-1}$ whenever $U_{i-1} \neq 0$, which holds for $1 \leq i \leq q-1$ since then $0 < i\theta < \pi$. This is (a).

For $1 \leq i \leq q-1$ we have $\sin(i\theta) > 0$ and $\sin((i+1)\theta) \geq 0$, with equality only at $i = q-1$. So $\beta_{q-1} = 0$, and $\beta_i > 0$ for $1 \leq i \leq q-2$. For $1 \leq i \leq q-2$,

$$\beta_i - \beta_{i+1} = \frac{U_i^2 - U_{i-1}U_{i+1}}{U_{i-1}U_i} = \frac{\sin^2\theta}{\sin(i\theta)\,\sin((i+1)\theta)} > 0,$$

by the identity $\sin^2 a - \sin(a-b)\sin(a+b) = \sin^2 b$ with $a = (i+1)\theta$ and $b = \theta$. Also $\beta_1 = \lambda$ and $\beta_{q-2} = \sin((q-1)\theta)/\sin((q-2)\theta) = \sin\theta/\sin 2\theta = 1/\lambda$. This is (b).

Since $U_{q-1} = 0$, $U_q = -1$ and $U_{q-2} = 1$, the matrix of τ^q is $-I$, so τ^q is the identity. Also $\tau(1/\lambda) = \lambda - \lambda = 0$. This is (c). For (d), note that τ is continuous and strictly increasing on $(0, \infty]$, and that $(\beta_{i+1}, \beta_i]$ lies in $(0, \infty]$ when $i+1 \leq q-2$. Since $\tau(\beta_j) = \beta_{j+1}$, the image is $(\beta_{i+2}, \beta_{i+1}]$. □

Lemma 3.2. *The map $G \circ F$ has order exactly q. For $m = 1, 2, 3$ the map $g_m \circ f$ has order 3, 4 and 6, and for $m \geq 4$ it has infinite order.*

Proof. Since $G \circ G$ is the identity, $G \circ F = G \circ \tau \circ G$, so $G \circ F$ has the same order as τ. By Lemma 3.1, τ^q is the identity, and $\tau^i(\infty) = \beta_i \neq \infty$ for $1 \leq i \leq q-1$. So τ has order q. For $m \leq 3$ the claim follows by Lemma 2.3. Now let $m \geq 4$. The matrix of $g_m \circ f(x) = -1/(m(x+1))$ is $\left(\begin{smallmatrix}0 & -1\\ m & m\end{smallmatrix}\right)$, and dividing by $\sqrt{m}$ gives a matrix N of determinant 1 and trace $\sqrt{m} \geq 2$. Its eigenvalues are μ and $1/\mu$ with $\mu + 1/\mu = \sqrt{m}$, so μ is real and positive. If $N^n = \pm I$ for some $n \geq 1$, then $\mu^n = \pm 1$, so $\mu = 1$ and $m = 4$. Then N has the double eigenvalue 1 but is not the identity, because its upper right entry is $-1/\sqrt{m}$. So N is conjugate to $\left(\begin{smallmatrix}1 & 1\\ 0 & 1\end{smallmatrix}\right)$, whose powers are never $\pm I$. Hence $g_m \circ f$ has infinite order. □

3.2 The parent map

Definition 3.3. The *parent map* $P\colon \mathbb{R} \setminus \{0\} \to \mathbb{R}$ is $P(X) = X - \lambda$ for $X > 0$ and $P(X) = -1/X$ for $X < 0$. Let $\mathcal{T}$ be the set of $X \in \mathbb{R}$ whose sequence $X, P(X), P^2(X), \ldots$ reaches 0, and for $X \in \mathcal{T}$ let $D(X)$ be the number of steps it takes. So $D(0) = 0$, and for $X \in \mathcal{T}$ with $X \neq 0$ we have $P(X) \in \mathcal{T}$ and $D(X) = D(P(X)) + 1$.

Note that for every $X \neq 0$ there is an edge $P(X) \to X$ in Γ_q. It is the F-edge from $X - \lambda$ when $X > 0$, and the G-edge from $-1/X \neq 0$ when $X < 0$. For $q = 3$ this is the parent map of [14, Definition 3.1].

The heart of [14] is the identity $d(x-1) = d(x) + 3$ for $x < 0$. The next lemma is its analogue. Its proof follows a path that winds once through the orbit of ∞.

Lemma 3.4 (the long in-edge)**.** *Let $Z < 0$ and $Y = -1/Z > 0$. Put $P_1 = Y + \lambda$, and for $1 \leq i \leq q-2$ put $N_i = -1/P_i$ and $P_{i+1} = N_i + \lambda = \tau(P_i)$. Put $N_{q-1} = -1/P_{q-1}$. Then every P_i is positive, every N_i is negative, $N_{q-1} = Z - \lambda$, and P maps each point of the path*

$$Y \xrightarrow{F} P_1 \xrightarrow{G} N_1 \xrightarrow{F} P_2 \xrightarrow{G} \cdots \xrightarrow{F} P_{q-1} \xrightarrow{G} N_{q-1}$$

to the point before it. Hence $Z \in \mathcal{T}$ if and only if $Z - \lambda \in \mathcal{T}$, and in that case $D(Z-\lambda) = D(Z) + 2q - 3$.

Proof. Since $P_1 > \lambda = \beta_1$, P_1 lies in $(\beta_1, \beta_0]$. By Lemma 3.1(d) and induction, $P_i \in (\beta_i, \beta_{i-1}]$ for $1 \le i \le q-1$. Since $\beta_{q-1} = 0$, every P_i is positive, so every N_i is defined and negative. The path applies $(GF)^{q-1}$ to Y. Now $G \circ F = G \circ \tau \circ G$, so $(GF)^q = G\tau^q G$ is the identity by Lemma 3.1(c). Hence $(GF)^{q-1} = (GF)^{-1} = F^{-1}G$, and $N_{q-1} = F^{-1}(G(Y)) = Z - \lambda$. The map G is applied only at the points P_i, which are not 0.

For the parent map, $P(N_i) = -1/N_i = P_i$ since $N_i < 0$, $P(P_{i+1}) = P_{i+1} - \lambda = N_i$ and $P(P_1) = Y$ since the P_i are positive. So the path has $2q-2$ edges and $P^{2q-2}(Z-\lambda) = Y = P(Z)$. If $Z \in \mathcal{T}$, then $Y \in \mathcal{T}$, so $Z - \lambda \in \mathcal{T}$ and $D(Z-\lambda) = 2q - 2 + D(Y) = 2q - 2 + D(Z) - 1$. Conversely, if $Z - \lambda \in \mathcal{T}$, then its parent sequence passes through Y, so $Y \in \mathcal{T}$, and $Z \in \mathcal{T}$ because $P(Z) = Y$. □

For $q = 3$ the path has four edges and this is (D4) of [14, Lemma 3.5]. For example, in the 2-tree take $z = -\frac{1}{2}$, so that $y = g_2(z) = 1$. The path is

$$1 \xrightarrow{f} 2 \xrightarrow{g_2} -\tfrac{1}{4} \xrightarrow{f} \tfrac{3}{4} \xrightarrow{g_2} -\tfrac{2}{3} \xrightarrow{f} \tfrac{1}{3} \xrightarrow{g_2} -\tfrac{3}{2} = z - 1,$$

and $d(-\frac{3}{2}) = d(-\frac{1}{2}) + 5 = 7$.

Proposition 3.5. *(a) $\mathcal{O} = \mathcal{T}$.*

(b) $\mathcal{T}$ is closed under F, under F^{-1}, and under G at nonzero points.

(c) $-\lambda \in \mathcal{T}$ and $D(-\lambda) = 2q - 4$.

(d) $\mathcal{O} \cup \{\infty\}$ is the orbit of 0 under the group generated by F and G.

Proof. First we show that $\mathcal{T}$ is closed under F and G.

G. Let $X \in \mathcal{T}$ with $X \neq 0$. If $X > 0$, then $G(X) < 0$ and $P(G(X)) = X$, so $G(X) \in \mathcal{T}$. If $X < 0$, then $G(X) = P(X) \in \mathcal{T}$.

F. Let $X \in \mathcal{T}$ and $W = X + \lambda$. If $W > 0$, then $P(W) = X$, so $W \in \mathcal{T}$. If $W = 0$, it is the root. If $W < 0$, then $X = W - \lambda$, and Lemma 3.4 with $Z = W$ gives $W \in \mathcal{T}$.

Now (a) follows. Since $0 \in \mathcal{T}$ and $\mathcal{T}$ is closed under both moves, $\mathcal{O} \subseteq \mathcal{T}$. For $X \in \mathcal{T}$, the reversed parent sequence of X is a path from 0 to X, so $\mathcal{T} \subseteq \mathcal{O}$.

For (c), note that $P(-\lambda) = 1/\lambda = \beta_{q-2}$. For $2 \le i \le q-2$ we have $\beta_i - \lambda = -1/\beta_{i-1} < 0$, so $P(\beta_i) = -1/\beta_{i-1}$ and $P(-1/\beta_{i-1}) = \beta_{i-1}$. Finally $P(\beta_1) = 0$. So the parent sequence of $-\lambda$ is

$$-\lambda,\ \beta_{q-2},\ -1/\beta_{q-3},\ \beta_{q-3},\ \ldots,\ -1/\beta_1,\ \beta_1,\ 0,$$

which has $1 + 2(q-3) + 1 = 2q - 4$ steps.

For F^{-1} in (b), let $X \in \mathcal{T}$. If $X > 0$, then $X - \lambda = P(X) \in \mathcal{T}$. If $X < 0$, use Lemma 3.4 with $Z = X$. If $X = 0$, use (c).

For (d), the set $\mathcal{O} \cup \{\infty\}$ is closed under F, F^{-1} and G, since F fixes ∞ and G swaps 0 and ∞. So it contains the orbit of 0. Conversely every point of $\mathcal{O}$ is the image of 0 under a word in F and G, and $\infty = G(0)$. □

3.3 The distance formula

Theorem 3.6. *For every $X \in \mathcal{O}$ we have $d(X) = D(X)$. Moreover, for $X \in \mathcal{O}$,*

(D1) if $X > 0$, then $D(X) = D(X - \lambda) + 1$,

(D2) if $X < 0$, then $D(X) = D(-1/X) + 1$,

(D3) *if* $X > 0$, *then* $D(-1/X) = D(X) + 1$,

(D4$_q$) *if* $X < 0$, *then* $D(X - \lambda) = D(X) + 2q - 3$.

Proof. All the points involved are in $\mathcal{T}$ by Proposition 3.5. (D1) and (D2) are the definition of D. (D3) is (D2) applied to $-1/X < 0$. (D4$_q$) is Lemma 3.4.

The reversed parent sequence of X is a path of length $D(X)$, so $d(X) \le D(X)$. For the reverse inequality we check that $D(W) \le D(V) + 1$ for every edge $V \to W$ with $V \in \mathcal{O}$. There are five cases.

- An F-edge into $W > 0$ comes from $V = W - \lambda$, and $D(W) = D(V) + 1$ by (D1).
- An F-edge into $W = 0$ gives $D(W) = 0$.
- An F-edge into $W < 0$ comes from $V = W - \lambda$, and $D(V) = D(W) + 2q - 3$ by (D4$_q$).
- A G-edge into $W < 0$ comes from $V = -1/W$, and $D(W) = D(V) + 1$ by (D2).
- A G-edge into $W > 0$ comes from $V = -1/W < 0$, and $D(V) = D(W) + 1$ by (D3).

A G-edge never ends at 0. Now take a shortest path $0 = V_0 \to V_1 \to \cdots \to V_n = X$. Then $D(V_i) \le D(V_{i-1}) + 1$ for each i, so $D(X) \le n = d(X)$. □

For $q = 3$ this is [14, Theorem 3.6]. The next corollary is Theorem C$_q$ of the introduction.

Corollary 3.7. *Let* $X \in \mathcal{O}$ *with* $X \neq 0$.

(a) The in-neighbors of X *are* $X - \lambda$ *and* $-1/X$, *and both lie in* $\mathcal{O}$. *If* $X > 0$, *their ranks are* $d(X) - 1$ *and* $d(X) + 1$. *If* $X < 0$, *they are* $d(X) + 2q - 3$ *and* $d(X) - 1$. *So* $P(X)$ *is the only in-neighbor of rank* $d(X) - 1$.

(b) X *has exactly one shortest word, which spells the reversed parent sequence. In the row construction of Kimberling [11], carried out with* F *and* G *as in [14, Section 2], each point is written down exactly once.*

(c) The last move of the shortest word is F *if* $X > 0$ *and* G *if* $X < 0$. *So a vertex is blue if and only if it is negative.*

(d) The root has the one child λ. *A vertex* $X > 0$ *has the two children* $X + \lambda$ *and* $-1/X$. *A vertex in* $(-\lambda, 0)$ *has the one child* $X + \lambda$, *and a vertex* $X \le -\lambda$ *is a leaf. Every child of* X *has rank* $d(X) + 1$. *For* $X < -\lambda$ *the value* $X + \lambda$ *appeared exactly* $2q - 3$ *ranks earlier, that is* $d(X + \lambda) = d(X) - (2q - 3)$, *and for* $X = -\lambda$ *it is the root.*

Here the *children* of X are the points Y with $P(Y) = X$.

Proof. (a) Only $X - \lambda$ is mapped to X by F, and only $-1/X$ by G. Both are in $\mathcal{O}$ by Proposition 3.5, and their ranks are given by Theorem 3.6. Since $2q - 3 \ge 3$, the two ranks differ.

(b) By induction on $d(X)$. The last edge of a shortest path comes from an in-neighbor of rank $d(X) - 1$, which is $P(X)$ by (a). For the rows, the proof of [14, Corollary 3.8(a)] applies word for word. The only extra point is that a parent c cannot reach X by both moves, since $c + \lambda = -1/c$ would give $c^2 + \lambda c + 1 = 0$, which has no real root because $\lambda < 2$.

(c) The edge $P(X) \to X$ is an F-edge for $X > 0$ and a G-edge for $X < 0$.

(d) We solve $P(Y) = X$. If $Y > 0$, then $Y = X + \lambda$, which is possible exactly when $X > -\lambda$. If $Y < 0$, then $X = -1/Y$, which is possible exactly when $X > 0$. Each child has rank $d(X) + 1$ by Theorem 3.6. For $X < -\lambda$, apply (D4$_q$) to $X + \lambda < 0$. □

In x-coordinates for $m = 2, 3$, a vertex $x > 0$ has the children $x + 1$ and $g_m(x)$, a vertex in $(-1, 0)$ has the child $x + 1$, and a vertex $x \le -1$ is a leaf. For $x < -1$ the value $x + 1$ appeared five ranks earlier when $m = 2$, and nine ranks earlier when $m = 3$.

Remark 3.8 (the groups). By Proposition 3.5(d), the λ_q-tree is the orbit of 0 under the Hecke group $H_q = \langle F, G \rangle$, with ∞ removed. It is known that H_q is isomorphic to the free product $C_2 * C_q$, with G and $G \circ F$ as generators [17, p. 1], [19, p. 4]. Im and Lee note that for $q = 4$ and $q = 6$ the Hecke group coincides with the group of all matrices $\left(\begin{smallmatrix} a & b\sqrt{N} \\ c\sqrt{N} & d \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} a\sqrt{N} & b \\ c & d\sqrt{N} \end{smallmatrix}\right)$ of determinant 1 with $a, b, c, d \in \mathbb{Z}$, where $N = 2$ and $N = 3$ [3, Section 2]. Conjugating by $X = \sqrt{m}\, x$ turns these into $\Gamma_0(m)$ together with its coset of the Fricke involution g_m. So for $m = 2, 3$ the maps f and g_m generate the Fricke group $\Gamma_0(m)^+$, whose orbit of ∞ is $\mathbb{Q} \cup \{\infty\}$ [1, Proposition 2], [19, p. 4]. Theorem D below says more, since the tree uses only forward moves. None of these facts is used in our proofs.

4 Every rational is reached when $m \le 3$

Theorem 4.1 (Theorem D)**.** *For $m = 1, 2, 3$ every rational number lies in the m-tree. For every rational x, the parent map $x \mapsto x - 1$ for $x > 0$ and $x \mapsto g_m(x)$ for $x < 0$ reaches 0 in exactly $d(x)$ steps.*

Proof. Let $\mathcal{T}_m$ be the set of points of the m-tree. By Lemma 2.3 and Proposition 3.5, $\mathcal{T}_m$ contains 0 and is closed under $x \mapsto x + 1$, under $x \mapsto x - 1$, and under g_m at nonzero points.

For a rational $x = u/v$ in lowest terms with $v \ge 1$ put

$$h(x) = \frac{v^2}{\gcd(v, m)}.$$

Since m is 1 or a prime, $\gcd(v, m)$ is 1 or m. Clearly $h(x + n) = h(x)$ for every integer n. We claim that

$$h(g_m(x)) = m\, x^2\, h(x) \qquad (x \ne 0).$$

If m does not divide v, then $g_m(x) = -v/(mu)$ is in lowest terms up to sign, since $\gcd(v, mu) = 1$. Its denominator $m|u|$ is divisible by m, so $h(g_m(x)) = m^2u^2/m = mu^2 = m\, x^2 v^2$. If m divides v, write $v = mv'$. Then $\gcd(u, m) = 1$, since $\gcd(u, v) = 1$ and m is 1 or a prime dividing v. So $g_m(x) = -v'/u$ is in lowest terms with a denominator prime to m, and $h(g_m(x)) = u^2 = m\, x^2\, (v^2/m)$. This proves the claim.

Since $\gcd(v, m)$ divides v^2, the value $h(x)$ is a positive integer, so we may use strong induction on $h(x)$. Given x, choose an integer n with $|x - n| \le \frac{1}{2}$ and put $y = x - n$. Then $x \in \mathcal{T}_m$ if and only if $y \in \mathcal{T}_m$, and $h(y) = h(x)$. If $y = 0$ we are done. Otherwise

$$h(g_m(y)) = m\, y^2\, h(y) \le \frac{m}{4}\, h(x) < h(x),$$

since $m \le 3$. By induction $g_m(y) \in \mathcal{T}_m$, and then $y = g_m(g_m(y)) \in \mathcal{T}_m$. The last sentence of the theorem is Theorem 3.6 together with Lemma 2.3. □

The bound $m/4 < 1$ is the only place where $m \le 3$ is used. For $m = 4$ the set $\mathcal{T}_4$ is not closed under $x \mapsto x - 1$, since it contains 0 but not -1 (Theorem 7.1). So the argument breaks down there, as it must.

5 The run rule and the rank formula

For $q = 3$, [14, Section 4] reads the rank off the negative continued fraction of x. Here we do the same with λ in place of 1. For integers $d_1, \dots, d_n$ write

$$[d_1, \dots, d_n]_\lambda = d_1\lambda - \cfrac{1}{d_2\lambda - \cfrac{1}{\ddots - \cfrac{1}{d_n\lambda}}},$$

whenever the denominators are nonzero. Its *tails* are $Y_j = [d_j, \dots, d_n]_\lambda$ for $1 \le j \le n$, together with $Y_{n+1} = \infty$, so that $Y_j = d_j\lambda - 1/Y_{j+1}$. Expansions of this kind with $\lambda = \lambda_q$ are Rosen's λ-continued fractions [15], as described in [16, Section 1.2] and [17]. Ours have all signs negative and all digits positive, as in [4]. At $q = 3$ this is the negative continued fraction $[d_1, \dots, d_n]^-$ of [14].

Lemma 5.1. *Let $d_1, \dots, d_n \ge 1$. All tails are defined and greater than $1/\lambda$ if and only if no $q-2$ consecutive digits equal 1. In that case every tail is positive.*

Proof. We use one observation. If $Y = \infty$ or $Y > 1/\lambda$, then $-1/Y \in (-\lambda, 0]$, so $d\lambda - 1/Y > (d-1)\lambda \ge \lambda$ for $d \ge 2$. Also $\lambda - 1/Y = \tau(Y)$.

Suppose first that no $q-2$ consecutive digits equal 1. We go from $j = n$ down to $j = 1$. Let i_j be the number of 1's at the start of $d_j, \dots, d_n$, so $i_j \le q-3$. We show that Y_j is defined and $Y_j \in (\beta_{i_j+1}, \beta_{i_j}]$. To start, $Y_{n+1} = \infty$ lies in $(\beta_1, \beta_0]$, and we treat it as having $i_{n+1} = 0$. Suppose the claim holds for Y_{j+1}, or $j = n$. If $d_j \ge 2$, then Y_{j+1} is ∞ or greater than $\beta_{i_{j+1}+1} \ge \beta_{q-2} = 1/\lambda$. So $Y_j > \lambda = \beta_1$ by the observation, and $i_j = 0$. If $d_j = 1$, then $i_j = i_{j+1} + 1$ and $Y_j = \tau(Y_{j+1}) \in (\beta_{i_j+1}, \beta_{i_j}]$ by Lemma 3.1(d), which applies since $i_{j+1} \le q-4$. In both cases $Y_j > \beta_{i_j+1} \ge \beta_{q-2} = 1/\lambda$, since $i_j + 1 \le q-2$.

Conversely, suppose all tails are defined and greater than $1/\lambda$, and let $d_a = \cdots = d_b = 1$ be a maximal run with $b - a + 1 \ge q-2$. Then $Y_{b+1} \in (\beta_1, \beta_0]$. Indeed, $Y_{b+1} = \infty$ if $b = n$, and otherwise $d_{b+1} \ge 2$ and $Y_{b+2} > 1/\lambda$, so $Y_{b+1} > \lambda$. By Lemma 3.1(d), $Y_{b+1-s} \in (\beta_{s+1}, \beta_s]$ for $0 \le s \le q-2$. At $s = q-2$ this gives $Y_{b+3-q} \in (0, 1/\lambda]$. Since $b + 3 - q \ge a$, this is a tail of our string, and it is at most $1/\lambda$, a contradiction. □

Lemma 5.2. *Every $X \in \mathcal{O}$ with $X > 0$ has exactly one expansion $X = [c_1, \dots, c_k]_\lambda$ with $k \ge 1$, all $c_j \ge 1$ and all tails $Y_j > 1/\lambda$ for $2 \le j \le k$. Its digits can be read off the parent sequence of X, and*

$$D(X) = c_1 + c_2 + \cdots + c_k + k - 1.$$

Proof. Existence. Let $c_1 = \lceil X/\lambda \rceil \ge 1$. The parent sequence of X subtracts λ exactly c_1 times and reaches $X - c_1\lambda \in (-\lambda, 0]$. If this is 0, then $X = c_1\lambda$ and $k = 1$. Otherwise one more step gives $Y_2 = -1/(X - c_1\lambda) > 1/\lambda$, which lies in $\mathcal{O}$ and is positive, and $X = c_1\lambda - 1/Y_2$. We repeat with Y_2 in place of X. Each round takes $c_j + 1$ steps of the parent sequence, except the last, which takes c_k steps. So the process stops, and counting steps gives $D(X) = c_1 + \cdots + c_k + k - 1$.

Uniqueness. Take any expansion of this kind. If $k \ge 2$, then $Y_2 > 1/\lambda$ gives $0 < 1/Y_2 < \lambda$, so $X \in ((c_1 - 1)\lambda, c_1\lambda)$. Hence X/λ is not an integer, $c_1 = \lceil X/\lambda \rceil$ and $Y_2 = 1/(c_1\lambda - X)$. If $k = 1$, then $X/\lambda = c_1$ is an integer. So the value of X decides whether $k = 1$, and it determines c_1 and Y_2. The tail $(c_2, \dots, c_k)$ is an expansion of Y_2 of the same kind, so induction on k finishes the proof. □

Recall from the introduction that a string $(c_1, \dots, c_k)$ with $k \ge 1$, $c_1 \ge 0$ and $c_2, \dots, c_k \ge 1$ is *reduced* if no $q-2$ consecutive digits among $c_2, \dots, c_k$ equal 1, except that a run of exactly $q-2$ ones starting at c_2 is allowed when $c_1 = 0$. Its word is $f^{c_1}gf^{c_2}g\cdots gf^{c_k}$ (with F and G in place of f and g for the λ_q-tree), and its weight is $c_1 + \cdots + c_k + k - 1$.

Theorem 5.3 (Theorem B_q)**.** *The map sending a reduced string to the value of its word at* 0 *is a bijection from the reduced strings onto* $\mathcal{O}$*. No reduced word applies* G *at* 0*, the rank of the value is the weight, and the reduced word of a point is its unique shortest word. The value of* $(c_1, \dots, c_k)$ *is* $[c_1, \dots, c_k]_\lambda$*, where* $0\lambda - 1/Y$ *means* $-1/Y$*. For* $m = 2, 3$*, every rational* x *has exactly one reduced string, with*

$$x = c_1 + g_m\Big(c_2 + g_m\big(c_3 + \cdots + g_m(c_k)\big)\Big) = c_1 - \cfrac{1}{mc_2 - \cfrac{1}{c_3 - \cfrac{1}{mc_4 - \cdots}}},$$

and $d(x) = c_1 + \cdots + c_k + k - 1$*.*

Proof. Applying the word to 0 from right to left gives $F^{c_k}(0) = c_k\lambda = Y_k$, then $-1/Y_k$, then Y_{k-1}, and so on. So the value is $[c_1, \dots, c_k]_\lambda$ whenever the tails $Y_2, \dots, Y_k$ are nonzero, and G is applied exactly at these tails.

Case $c_1 \geq 1$. By Lemma 5.1 applied to $(c_2, \dots, c_k)$, the run rule holds if and only if the tails Y_j, $j \geq 2$, are defined and greater than $1/\lambda$. Then G is applied only at positive points, and the value X is greater than $(c_1 - 1)\lambda \geq 0$. So X is a positive point of $\mathcal{O}$, and by the uniqueness in Lemma 5.2 its expansion is $(c_1, \dots, c_k)$, with $D(X)$ equal to the weight. Conversely, every positive $X \in \mathcal{O}$ arises in this way by Lemma 5.2. So the reduced strings with $c_1 \geq 1$ match the positive points of $\mathcal{O}$ one to one, and the rank is the weight by Theorem 3.6.

Case $c_1 = 0$*,* $k \geq 2$*.* A run of 1's starting at c_2 of length L leaves a run of length $L - 1$ at the start of $c_3, \dots, c_k$. So the rule says exactly that $(c_2, \dots, c_k)$ is a reduced string with first digit $c_2 \geq 1$. By the first case it is the string of a unique positive $Y \in \mathcal{O}$ with $d(Y) = c_2 + \cdots + c_k + k - 2$. The value of $(0, c_2, \dots, c_k)$ is $G(Y) = -1/Y$. The map G is a bijection from the positive points of $\mathcal{O}$ onto the negative ones (Proposition 3.5), and $d(-1/Y) = d(Y) + 1$ by (D3), which is the weight.

Case $c_1 = 0$*,* $k = 1$*.* This is the empty word, with value 0.

The three cases give the positive points, the negative points and 0, each exactly once. In each case the reduced word spells the reversed parent sequence (Lemma 5.2 and (D2)), which is the unique shortest word by Corollary 3.7(b). For $m = 2, 3$, Theorem 4.1 and Lemma 2.3 turn this into a statement about all of $\mathbb{Q}$, and $X = \sqrt{m}\,x$ turns $[c_1, \dots, c_k]_\lambda$ into the expression in g_m. □

At $q = 3$ the rule says $c_2, \dots, c_k \geq 2$, except that $c_2 = 1$ is allowed when $c_1 = 0$, and Theorem 5.3 is [14, Lemma 4.1 and Theorem 4.2]. The next corollary is the analogue of [14, Corollary 4.3].

Corollary 5.4. *Let* $n \geq 1$*. The positive points of rank* n *match the compositions* $n = s_1 + \cdots + s_k$ *with* $s_1 \geq 1$ *and* $s_2, \dots, s_k \geq 2$ *in which no* $q - 2$ *consecutive parts among* $s_2, \dots, s_k$ *equal* 2*, via* $s_1 = c_1$ *and* $s_j = c_j + 1$ *for* $j \geq 2$*. The negative points of rank* n *are the* $-1/Y$ *for the positive points* Y *of rank* $n - 1$*.*

Proof. The weight of a string is $c_1 + (c_2 + 1) + \cdots + (c_k + 1)$. Now apply Theorem 5.3 and (D2), (D3). □

Example 5.5. Take $m = 2$, so $q = 4$, and $n = 5$. The compositions of 5 with $s_1 \geq 1$ and later parts at least 2 are 5, $1 + 4$, $2 + 3$, $3 + 2$ and $1 + 2 + 2$. The last one has two consecutive parts equal to 2 after the first part, so it is not allowed. The other four give the strings (5), (1, 3), (2, 2) and (3, 1), with values 5, $1 + g_2(3) = \frac{5}{6}$, $2 + g_2(2) = \frac{7}{4}$ and $3 + g_2(1) = \frac{5}{2}$. The positive points of rank 4 are 4, $\frac{3}{4}$ and $\frac{3}{2}$, and their images under g_2 are $-\frac{1}{8}$, $-\frac{2}{3}$ and $-\frac{1}{3}$. Together these are the seven rationals of rank 5 in Table 2.

Remark 5.6 (where the run rule comes from). The proofs above do not need the following explanation. By Lemma 3.2, $(G \circ F)^{q-1} = (G \circ F)^{-1} = F^{-1} \circ G$. Hence, for $a, b \geq 1$,

$$F^a\, G\, (FG)^{q-2}\, F^b = F^{a-1}\, G\, F^{b-1}$$

wherever both sides are defined. The left side is the word of a string with an interior run of $q-2$ digits equal to 1, between the digits a and b. The right side is $2q-2$ moves shorter. So such a word is never a shortest word, and this is the run rule. At $q = 3$ it is the relation $fgfgf = g$ used in [14, Remark 3.9] and [9]. The other rearrangements $(GF)^j G = (F^{-1}G)^{q-j} G$ of the relation put $q-j$ copies of F^{-1} into the word. Heuristically, a path can absorb an F^{-1} only into the blocks at the two ends of the run, so only $q - j = 2$ gives a shortcut of this kind. This suggests why shorter runs of 1's survive, and Theorem 5.3 proves it.

The exception at the root has the same source. By Proposition 3.5(c), $(GF)^{q-2}(0) = -\lambda = F^{-1}(0)$, so $F^a(GF)^{q-2}(0) = F^{a-1}(0)$ for $a \geq 1$. A run of $q-2$ ones at c_2 followed by $c_1 \geq 1$ can therefore be shortened, while the word $(GF)^{q-2}$ itself, with $c_1 = 0$, is the shortest word for $-\lambda$.

In the language of the free product $C_2 * C_q$ of Remark 3.8, put $u = G \circ F$, so $F = G \circ u$. Then an interior run of j digits equal to 1 becomes the syllable u^{j+2}, which is trivial exactly when $j \equiv -2 \pmod q$. The shortest such run has $j = q-2$. One could prove Theorem 5.3 from the normal form theorem for free products, but the elementary proof above avoids the group theory.

6 Counting the ranks

Recall that

$$D_q(t) = 1 - t - t^3 - \cdots - t^{2q-3}, \qquad R_q(t) = 1 + t^2 + \cdots + t^{2q-6}, \qquad N_q(t) = 1 + t^2 + \cdots + t^{2q-4} - t^{2q-3},$$

with $R_3 = 1$. Note that $t^3 R_q(t) = t^3 + t^5 + \cdots + t^{2q-3}$, so $D_q = 1 - t - t^3 R_q$.

Theorem 6.1 (Theorem A_q)**.** *Let $q \geq 3$. Then*

$$\sum_{n \geq 1} r_q(n)\, t^n = \frac{t\, R_q(t)}{D_q(t)}, \qquad \sum_{n \geq 0} a_q(n)\, t^n = \frac{N_q(t)}{D_q(t)},$$

$a_q(0) = 1$ and $a_q(n) = r_q(n) + r_q(n-1)$ for $n \geq 1$, and $\gcd(N_q, D_q) = 1$. Hence $a_q(n) = a_q(n-1) + a_q(n-3) + \cdots + a_q(n-2q+3)$ for every $n \geq 2q-2$, and at $n = 2q-3$ the left side minus the right side is -1. For $m = 1, 2, 3$ these are the counts of the rationals of rank n in the m-tree.

Proof. Positive points. By Theorem 5.3, $r_q(n)$ is the number of strings with $c_1 \geq 1$, $c_2, \ldots, c_k \geq 1$, no run of more than $q-3$ ones among $c_2, \ldots, c_k$, and weight n. Give c_1 the weight c_1 and each c_j with $j \geq 2$ the weight $c_j + 1$, which counts the g next to it. Then c_1 contributes $t/(1-t)$, a later digit 1 contributes $u = t^2$, and a later digit at least 2 contributes $B = \sum_{c \geq 2} t^{c+1} = t^3/(1-t)$. Read from left to right, the sequence $c_2, \ldots, c_k$ is in exactly one way a run of at most $q-3$ ones followed by some number of blocks, where each block is one digit at least 2 followed by a run of at most $q-3$ ones. A run contributes $1 + u + \cdots + u^{q-3} = R_q(t)$. So

$$\sum_{n \geq 1} r_q(n)\, t^n = \frac{t}{1-t} \cdot R_q \cdot \frac{1}{1 - BR_q} = \frac{tR_q}{(1-t)(1-BR_q)} = \frac{tR_q}{1 - t - t^3 R_q} = \frac{tR_q}{D_q}.$$

The geometric series makes sense as a formal power series because B has no constant term.

Negative points. By (D2) and (D3), $X \mapsto -1/X$ is a bijection from the negative points of rank n onto the positive points of rank $n-1$. Every point of rank $n \geq 1$ is nonzero, so $a_q(n) = r_q(n)+r_q(n-1)$. Hence

$$\sum_{n\geq 0} a_q(n)\, t^n = 1 + (1+t)\frac{tR_q}{D_q} = \frac{D_q + (t + t^2 + t^3 + \cdots + t^{2q-4})}{D_q} = \frac{N_q}{D_q},$$

since $(1+t)tR_q = t + t^2 + \cdots + t^{2q-4}$, and adding this to D_q cancels $t, t^3, \ldots, t^{2q-5}$.

Lowest terms. Let α be a common complex root of N_q and D_q. Then $\alpha \neq 0$ since $D_q(0) = 1$, and $\alpha \neq 1$ since $D_q(1) = 2-q$. Now $N_q - D_q = t+t^2+\cdots+t^{2q-4}$ vanishes at α, so $\alpha^{2q-4} = 1$. Also

$$(1-t^2)\, D_q(t) = 1 - t - t^2 + t^{2q-1},$$

and at α the right side is $1-\alpha-\alpha^2+\alpha^3 = (1-\alpha)^2(1+\alpha)$. So $\alpha = -1$. But $D_q(-1) = 1+1+(q-2) = q \neq 0$, a contradiction.

The recurrence. The coefficient of t^n in $D_q(t)\sum_k a_q(k)t^k$ is $a_q(n) - a_q(n-1) - a_q(n-3) - \cdots - a_q(n-2q+3)$, with $a_q(j) = 0$ for $j < 0$. It equals the coefficient of t^n in N_q, which is 0 for $n \geq 2q-2$ and -1 for $n = 2q-3$. The last sentence of the theorem is Lemma 2.3. □

At $q = 3$ this is [14, Theorem 5.1]. The defect -1 at $n = 2q-3$ comes from the root, as it does there, since the word $F(GF)^{q-2}$ takes the root back to itself (Remark 5.6).

For $m = 2$ the positive counts have generating function $(t+t^3)/D_4$. The OEIS entry A060961, which counts compositions into parts 1, 3 and 5, has generating function $1/D_4$ and offset 0 [10]. So

$$r(n) = \text{A060961}(n-1) + \text{A060961}(n-3) \qquad (m = 2,\ n \geq 1),$$

with A060961$(j) = 0$ for $j < 0$. We found no OEIS entry for the sequences $a(n)$ with $m = 2$ or $m = 3$, for the $q = 5$ counts, or for the coefficients of $1/D_6$.

Proposition 6.2 (growth)**.** *Let $q \geq 3$.*

(a) D_q has exactly one root t_0 in $(0,1)$. It is simple, and every other complex root of D_q has modulus greater than 1. So $\psi_q = 1/t_0$ is a Pisot number. It is the largest real root of $\psi^{2q-3} = \psi^{2q-4} + \psi^{2q-6} + \cdots + \psi^2 + 1$.

(b) $a_q(n) = C_q\psi_q^n + \varepsilon_q(n)$, where

$$C_q = \frac{(1+t_0)\, R_q(t_0)}{-D_q'(t_0)} > 0$$

and $\varepsilon_q(n) \to 0$ exponentially fast. The share of negative (blue) points among the points of rank n tends to $1/(1+\psi_q)$.

(c) $\psi_3 < \psi_4 < \psi_5 < \cdots$, and ψ_q tends to the golden ratio $\varphi = (1+\sqrt{5})/2$.

Proof. (a) On $[0,1]$ the sum $t + t^3 + \cdots + t^{2q-3}$ increases from 0 to $q-1 \geq 2$, so D_q has exactly one root t_0 in $(0,1)$, and $D_q'(t_0) < 0$. Put $n = 2q-1$ and $H(t) = (1-t^2)D_q(t) = 1 - t - t^2 + t^n$. For $t = \rho e^{i\phi}$ and $c = \cos\phi$ a direct expansion gives

$$\left|1 - t - t^2\right|^2 = 1 + 3\rho^2 + \rho^4 - 2\rho c\,(1-\rho^2) - 4\rho^2 c^2.$$

This is concave in c, so on the circle $|t| = \rho$ its minimum is at $c = 1$ or $c = -1$, where it equals $(\rho^2+\rho-1)^2$ or $(1+\rho-\rho^2)^2$. For $0.62 < \rho \leq 1$ the smaller of the two is $(\rho^2+\rho-1)^2$, and $\rho^2+\rho-1 > 0$. The function $\rho^2+\rho-1-\rho^n$ vanishes at $\rho = 1$ and has derivative $3-n < 0$ there,

so it is positive on some interval $(1-\delta, 1)$. For ρ in this interval with $\rho > \max(t_0, 0.62)$ we get $|t^n| < |1-t-t^2|$ on $|t| = \rho$. By Rouché's theorem H has as many zeros in $|t| < \rho$ as $1-t-t^2$, whose zeros are $0.618\ldots$ and $-1.618\ldots$. That is one zero, and it must be t_0. Letting $\rho \to 1$, we see that t_0 is the only zero of D_q in the open unit disk. On $|t| = 1$ a zero of H needs $|1-t-t^2| = 1$, that is $5 - 4c^2 = 1$, so $t = \pm 1$. But $D_q(1) = 2 - q$ and $D_q(-1) = q$ are nonzero. So every other root of D_q has modulus greater than 1. The reversed polynomial $t^{2q-3}D_q(1/t) = t^{2q-3} - t^{2q-4} - \cdots - 1$ is monic with integer coefficients, and ψ_q is its only root of modulus at least 1. The conjugates of ψ_q are among its roots, so ψ_q is a Pisot number.

(b) By partial fractions, $a_q(n) = Ct_0^{-n} + \sum_i p_i(n)\, t_i^{-n}$ for large n, where the t_i are the other roots of D_q and the p_i are polynomials. These terms tend to 0 exponentially since $|t_i| > 1$. The simple pole at t_0 gives $C = -N_q(t_0)/(t_0 D_q'(t_0))$. Since $N_q = D_q + (1+t)tR_q$, we have $N_q(t_0) = (1+t_0)t_0 R_q(t_0) > 0$, which gives $C = C_q$. In the same way $r_q(n)$ is asymptotic to $R_q(t_0)\psi_q^n/(-D_q'(t_0))$. The negative points of rank n are counted by $r_q(n-1)$, so their share tends to $t_0/(1+t_0) = 1/(1+\psi_q)$.

(c) Since $D_{q+1} = D_q - t^{2q-1}$, we get $D_{q+1}(t_0) < 0$, so the root of D_{q+1} in $(0,1)$ is smaller than t_0. Hence ψ_q increases with q. Put $L(t) = (1-t-t^2)/(1-t^2)$. Then $D_q(t) - L(t) = t^{2q-1}/(1-t^2) > 0$ on $(0,1)$. Since $L(1/\varphi) = 0$, we get $D_q(1/\varphi) > 0$ and so $t_0 > 1/\varphi$. Now let $0 < \epsilon < 1 - 1/\varphi$ and $s = 1/\varphi + \epsilon$. Then $L(s) < 0$, and $D_q(s) - L(s) = s^{2q-1}/(1-s^2)$ tends to 0 as $q \to \infty$. So $D_q(s) < 0$ for large q, which gives $t_0 < s$. Hence $t_0 \to 1/\varphi$. □

Table 3 lists the constants, and Figure 2 shows $a_q(n)/\psi_q^n$ settling down to C_q. Note that C_q is not monotone in q. Still C_q tends to the constant $C_\infty = \varphi/\sqrt{5}$ of the free case of Section 7, where $a(n) = F_{n+1}$. To see this, put $s = 1/\varphi$. By part (c) the roots t_0 decrease to s, and $t_0 \le 0.69$ for every q. On $[0, 0.7]$ the polynomials R_q and D_q' converge uniformly to $1/(1-t^2)$ and $-(1+t^2)/(1-t^2)^2$, since they are partial sums of power series with radius 1. Hence

$$C_q \to \frac{(1+s)(1-s^2)}{1+s^2} = \frac{1}{1+s^2} = \frac{1}{3-\varphi} = \frac{\varphi}{\sqrt{5}},$$

where we used $s^2 = 1 - s$ twice and $\varphi(3-\varphi) = 2\varphi - 1 = \sqrt{5}$. The constant ψ_4 is A293506 in the OEIS [12].

q	m	ψ_q	C_q	$1/(1+\psi_q)$
3	1	1.4655712319	0.7019310679	0.4055855240
4	2	1.5701473122	0.7569789423	0.3890827562
5		1.6013473338	0.7487899136	0.3844161781
6	3	1.6119303966	0.7378594370	0.3828585943
∞	≥ 4	1.6180339887	0.7236067977	0.3819660113

Table 3: The growth rate ψ_q, the constant C_q of Proposition 6.2, and the limiting share of negative points. All digits shown are correct.

Remark 6.3. For $q = 3$, [14, Proposition 5.2] gives the error term $O(\psi^{-n/2})$. For $q \ge 4$ the other roots of D_q are only a little outside the unit circle (the smallest moduli are 1.0999, 1.0556 and 1.0343 for $q = 4, 5, 6$), so the error term decays slowly. We checked with 40-digit arithmetic that $a_q(n)$ is the nearest integer to $C_q\psi_q^n$ for $1 \le n \le 80$ and $q = 3, 4, 5, 6$. For $q = 3, 4, 6$ we also bounded the error by $E(n) = \sum_i |c_i|\, |t_i|^{-n}$, where the c_i are the partial fraction coefficients at the other roots t_i of D_q. This bound decreases in n, and it is below $\frac{1}{2}$ for $n \ge 3$, $n \ge 6$ and $n \ge 11$ when $q = 3$, 4 and 6. Together with the direct check, this gives the rounding for every $n \ge 1$ when $q = 3, 4, 6$. These are computer checks in 40-digit floating point, not in interval arithmetic, so they are not proofs.

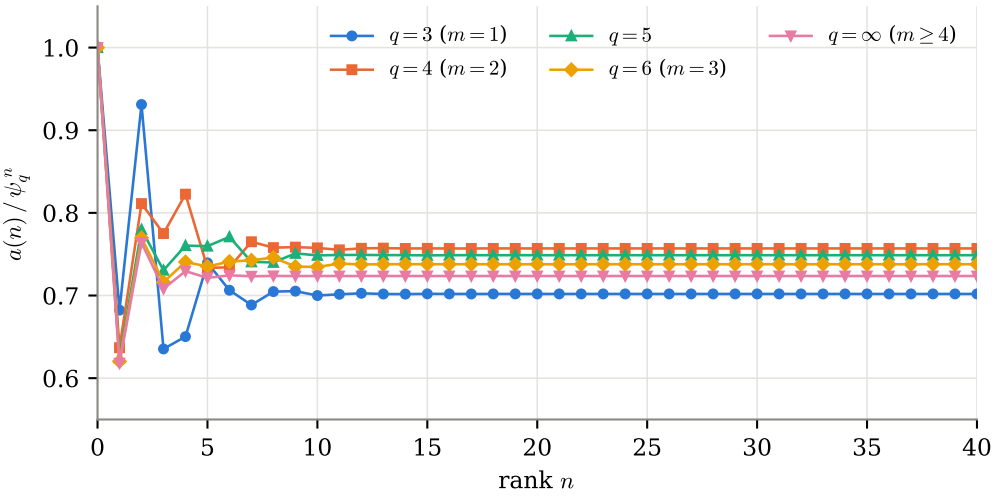


Figure 2: The normalized counts $a_q(n)/\psi_q^n$ for $0 \le n \le 40$. Each sequence tends to the constant C_q of Table 3.

7 The free case $m \geq 4$

For $m \geq 4$ the map $g_m \circ f$ has infinite order (Lemma 3.2), so there is no relation to shorten words, and every string is reduced. We work directly in x-coordinates. Call a string $(c_1, \ldots, c_k)$ with $k \geq 1$, $c_1 \geq 0$ and $c_2, \ldots, c_k \geq 1$ a *free string*. Its value is

$$x = c_1 + g_m\Big(c_2 + g_m\big(c_3 + \cdots + g_m(c_k)\big)\Big),$$

its tails are $T_j = c_j + g_m(T_{j+1})$ with $T_k = c_k$, and its weight is $c_1 + \cdots + c_k + k - 1$.

Theorem 7.1 (Theorem E)**.** *Let $m \geq 4$, $f(x) = x + 1$ and $g_m(x) = -1/(mx)$.*

(a) Every tail T_j with $j \geq 2$ of a free string is greater than $\frac{1}{2}$, so g_m is never applied at 0. A free string with $c_1 \geq 1$ has value in $(c_1 - 2/m,\, c_1]$, and one with $c_1 = 0$ and $k \geq 2$ has value in $(-2/m,\, 0)$. Distinct free strings have distinct values.

(b) The points of the m-tree are exactly the values of free strings, and the rank of a point is the weight of its string.

(c) $a(n) = F_{n+1}$ for every $n \geq 0$, and the number of positive points of rank $n \geq 1$ is F_n. The generating function is $1/(1 - t - t^2)$.

(d) A vertex is blue if and only if it is negative, and every point has exactly one shortest word.

(e) No rational in $(-\infty, -2/m]$ is in the tree, and for each integer $c \geq 1$ no rational in $(c - 1,\, c - 2/m]$ is in the tree. In particular -1 and $\frac{1}{2}$ are never reached.

Proof. (a) First $T_k = c_k \geq 1 > \frac{1}{2}$. If $T > \frac{1}{2}$ and $c \geq 1$, then $c + g_m(T) = c - 1/(mT) > 1 - 2/m \geq \frac{1}{2}$. By downward induction every tail T_j with $j \geq 2$ is greater than $\frac{1}{2}$, and in particular g_m is applied only at positive numbers. For $k \geq 2$ the value is $c_1 - 1/(mT_2)$ with $0 < 1/(mT_2) < 2/m$, which gives the two intervals. For $k = 1$ the value is c_1. Since $2/m \leq \frac{1}{2} < 1$, the intervals $(c - 2/m, c]$ for

$c = 1, 2, \ldots$ and $(-2/m, 0)$ are disjoint, and none contains 0. So the value x decides whether $c_1 = 0$, and if $c_1 \geq 1$ then $c_1 = \lceil x \rceil$, and $k = 1$ exactly when x is an integer. When $k \geq 2$ the tail value T_2 is $g_m(x - c_1)$. So the value determines c_1, whether $k = 1$, and T_2, and induction on k shows that distinct free strings have distinct values. The string (0) has the value 0.

(b) Let V be the set of values and $W(x)$ the weight of the string of $x \in V$. Each string is a word of length $W(x)$ that reaches x, so $d \leq W$ on V. Next, V is closed under the moves, and each move raises W by at most 1. Indeed, $f(0) = 1$ has the string (1). If $y > 0$ has the string $(c_1, \ldots, c_k)$, then $f(y)$ has the string $(c_1 + 1, c_2, \ldots, c_k)$ and $g_m(y)$ has the string $(0, c_1, \ldots, c_k)$, both of weight $W(y) + 1$. If $y < 0$ has the string $(0, c_2, \ldots, c_k)$, then $g_m(y)$ is the value of $(c_2, \ldots, c_k)$, of weight $W(y) - 1$, and $f(y)$ has the string $(1, c_2, \ldots, c_k)$, of weight $W(y) + 1$. Since $W(0) = 0$, every point reached by n moves lies in V and has $W \leq n$. So the tree is V and $d = W$.

(c) For the positive points, $c_1 \geq 1$ contributes $t/(1 - t)$, and each later digit $c \geq 1$ together with its g_m contributes t^{c+1}, which sums to $t^2/(1 - t)$. So

$$\sum_{n \geq 1} r(n)\, t^n = \frac{t}{1 - t} \cdot \frac{1}{1 - t^2/(1 - t)} = \frac{t}{1 - t - t^2}.$$

The map g_m is a bijection from the positive points of rank $n-1$ onto the negative points of rank n, by (b), since $g_m(y)$ has the string $(0, c_1, \ldots, c_k)$. So $\sum_n a(n)t^n = 1 + (1+t)t/(1-t-t^2) = 1/(1-t-t^2)$.

(d) Let $x < 0$ be in the tree. Its in-neighbors are $x - 1$ and $g_m(x)$. Since $x - 1 < -1 \leq -2/m$, the point $x - 1$ is not in the tree by (a) and (b). So the last move of every path to x is g_m. Let $x > 0$. Its in-neighbors are $x - 1$ and $g_m(x)$, and $g_m(x)$ has rank $d(x) + 1$ by the proof of (c). So the last move of a shortest path to x is f. Uniqueness of the shortest word follows by induction on the rank.

(e) This is (a) and (b). Note that $-1 \leq -2/m$, and $\frac{1}{2} \in (0, 1 - 2/m]$ since $m \geq 4$. □

For $m = 4$ the parent map of Section 3 does not terminate everywhere. It sends $\frac{1}{2}$ to $-\frac{1}{2}$ and $-\frac{1}{2}$ back to $g_4(-\frac{1}{2}) = \frac{1}{2}$, and both points are outside the tree by (e). With $x = ky$, the tree of $x \mapsto x + k$ and $x \mapsto -1/x$ is the k^2-tree, so Theorem E applies to it for every $k \geq 2$.

8 Computational verification

The proofs above do not depend on any computation. As a check, the script `verify_general.py` (Python with the standard `fractions` module, `sympy` for the transfer matrix, and `mpmath` with 40 digits for the constants of Table 3, about five minutes) recomputes every number in this paper by two independent methods. The first is breadth-first search from 0 in exact arithmetic, which is Kimberling's row construction. The second enumerates the reduced strings of Theorem 5.3 and evaluates each word at 0. Rationals are Python fractions, and the λ_5-tree uses exact arithmetic in $\mathbb{Q}(\sqrt{5})$ with $\lambda_5 = (1 + \sqrt{5})/2$. The script checks the following, and none of the checks fails.

1. For $m = 1$ to rank 24 (21,304 points), $m = 2$ to rank 28 (638,804 points), $m = 3$ to rank 30 (3,228,541 points), $q = 5$ to rank 20 (24,514 points) and $m = 4, 5, 6$ to rank 22 (75,024 points each), the two methods give the same points with the same ranks. No reduced word applies g at 0, no two reduced strings have the same value, and every point found by the search has a reduced string. The counts equal the coefficients of N_q/D_q, or of $1/(1 - t - t^2)$ in the free case.
2. In the same ranges, the recurrence of Theorem 6.1 holds from $n = 2q - 2$ on and fails at $n = 2q - 3$ with defect -1, and $a(n) = r(n) + r(n - 1)$.

3. At every point found by the search, the parent map, run from the value alone, reaches 0 in exactly $d(x)$ steps, and the digits it reads off equal the reduced string. Each point is reached at its rank by exactly one edge, and it is reached by g exactly when it is negative. For $x < -1$ in range, $d(x+1) = d(x) - (2q-3)$, and $d(-1) = 2q-4$.
4. For $m = 1, 2, 3$ the parent map reaches 0 from all 4,407 rationals u/v with $|u|, v \le 60$, and for $m = 4$ it fails from 3,726 of them. The identities $h(x+1) = h(x)$ and $h(g_m(x)) = mx^2 h(x)$ of the proof of Theorem 4.1 hold for $|u|, v \le 200$. For $m = 4, 5, 6$ the points lie in the intervals of Theorem 7.1(a), and -1 and $\frac{1}{2}$ are not reached.
5. A transfer matrix for the run rule, solved symbolically, gives the generating function N_q/D_q with $\gcd(N_q, D_q) = 1$ for $3 \le q \le 8$, without using the formula of Theorem 6.1.
6. The constants of Table 3 agree when computed from the residue $-N_q(t_0)/(t_0 D_q'(t_0))$, from the formula of Proposition 6.2, and from $a_q(400)/\psi_q^{400}$. Every other root of D_q has modulus greater than 1. Remark 6.3 is checked here.
7. Table 2, the examples, the identity with A060961 for $1 \le n \le 21$, and the numbers of vertices in Figure 1 (31 and 48).

Table 4 lists the first counts. The script `make_figures.py` draws Figures 1 and 2.

n	$q=3$ $(m=1)$	$q=4$ $(m=2)$	$q=5$	$q=6$ $(m=3)$	$q=\infty$ $(m\ge 4)$
0	1	1	1	1	1
1	1	1	1	1	1
2	2	2	2	2	2
3	2	3	3	3	3
4	3	5	5	5	5
5	5	7	8	8	8
6	7	11	13	13	13
7	10	18	20	21	21
8	15	28	32	34	34
9	22	44	52	54	55
10	32	69	83	87	89
11	47	108	133	141	144
12	69	170	213	227	233
13	101	267	341	366	377
14	148	419	546	590	610
15	217	658	874	951	987
16	318	1,033	1,400	1,533	1,597
17	466	1,622	2,242	2,471	2,584
18	683	2,547	3,590	3,983	4,181
19	1,001	3,999	5,749	6,420	6,765
20	1,467	6,279	9,206	10,349	10,946

Table 4: The counts $a_q(n)$ for $0 \le n \le 20$. The columns for $q = 3, 4, 6$ count rationals, the column for $q = 5$ counts points of $\mathbb{Q}(\sqrt{5})$, and the last column is the Fibonacci numbers F_{n+1}.

9 Related work and open questions

Related work. The groups $H_q = \langle F, G \rangle$ were defined by Hecke [2], and the continued fractions of Section 5 go back to Rosen [15]. The full texts of these two papers were not available to us,

and we rely on the later sources cited here for their content. Rosen's fractions, as described in [16, Section 1.2] and [17], use the nearest multiple of λ and digits of both signs, so his normal form is not ours. The closest work we know is by Short and Walker [17]. They show that Rosen fractions correspond to paths in a Farey-type graph and characterize the *geodesic* expansions, those with the fewest terms, by forbidden blocks of digits [17, Theorem 1.3 for even q and Theorem 7.1 for odd q]. Their condition and our run rule both come from the relation $(GF)^q = \mathrm{id}$. But they count terms and allow digits of both signs, while we count moves and use only forward moves, so the forbidden blocks differ. For example, for $q = 6$ they forbid three consecutive digits 1 and we forbid four. Janvresse, Rittaud and de la Rue study the positive all-minus λ-fractions of Section 5 as a dynamical system conjugate to a β-shift [4]. Their count is by number of digits, which gives a different sequence. Taha builds a Stern–Brocot process for the Hecke groups [18], which inserts mediants and does not count by word length.

For free products there is a classical formula for growth series [13], but it needs a generating set that is a union of generating sets of the factors. Our generators f and g are not of that form, since $f = g \circ (g \circ f)$ mixes the two factors. Also we use only forward moves, and we count points, not group elements.

Kimberling and Moses [7] studied several trees of this kind. Their Theorem 3.2 shows that the tree of $x \mapsto x + k$ and $x \mapsto k/x$ with root 1 has Fibonacci counts. Their Example 4.2 states the recurrence $a(n) = a(n-1) + a(n-3)$ for the tree of $x+1$ and $-1/x$ with root 1. Their Example 4.4, with $x \mapsto x + 1$ and $x \mapsto -m/x$, is not in our family, since there the composite has trace squared over determinant equal to $1/m$, which is not $4\cos^2(\pi/q)$ for any q.

As far as we know, the counts $a(n)$ for $m = 2$ and $m = 3$, the run rule, Theorem D, and the fact that the rank generating function depends only on q are new. The group theory around them is classical.

The cusp set. By Proposition 3.5(d), the λ_q-tree is the orbit of ∞ under H_q with ∞ removed, since $G(\infty) = 0$. This orbit is the set of cusps of H_q. For $q = 5, 8, 10, 12$, Leutbecher showed that the cusps are $\lambda_q \mathbb{Q}(\lambda_q^2)$ together with ∞ (Satz 2, as stated in the introduction of [8]). Winsor also attributes this case to Leutbecher and McMullen [19]. Hence the λ_5-tree is all of $\mathbb{Q}(\sqrt{5})$. For most $q \geq 7$ the cusp set is not known [16, 19]. Our Theorems A_q, B_q and C_q hold for every q all the same, because they never use the cusp set.

Open questions.

1. *Other involutions.* Any rational Möbius involution $g(x) = (ax + b)/(cx - a)$ can be paired with $f(x) = x + 1$. Then $g \circ f$ has trace c and determinant $-(a^2 + bc)$, and it has finite order q exactly when $c^2/(-(a^2 + bc))$ is 0, 1, 2 or 3, with $q = 2, 3, 4, 6$, assuming $-(a^2 + bc) > 0$. Is the rank generating function then always N_q/D_q, perhaps up to a correction at the root? A fixed point of g or a pole inside the tree may change the small ranks.

2. *The free case.* Describe exactly which rationals the m-tree reaches for $m \geq 4$. Theorem 7.1 gives only necessary conditions.

3. *Rounding.* Prove that $a_q(n)$ is the nearest integer to $C_q \psi_q^n$ for every $n \geq 1$ and every q, or give a rigorous proof for $q = 4, 6$ (Remark 6.3).

4. *Ordinary continued fractions.* For $q = 3$ the rank also has a formula in terms of the ordinary continued fraction [14, Theorem 4.8]. Is there a similar formula for $m = 2$ and $m = 3$?

5. *The constants.* Is there a closed form for the share of negative points, or for C_q, that explains why C_q is not monotone in q?

Data and code availability

The scripts `verify_general.py` and `make_figures.py` are in the repository https://github.com/apemm/Kagey-Problems, in the folder `kagey-problems/problem137/general`.

Acknowledgment. I thank Peter Kagey for posing the problem.

A note on the use of AI

The following ideas were suggested by an AI model (Claude, made by Anthropic, model Opus 5.5, used through Claude Code in October 2026). These are the generalization to $x \mapsto -1/(mx)$, the reading of the 3 in Problem 137 as the order of $g \circ f$, the run rule of Theorem B_q, the reduction of $x \mapsto x + k$ with $x \mapsto -1/x$ to the case $m = k^2$, the title, the directions on $q = 5$ and on the growth constants ψ_q and C_q, the question on other involutions, and a proof route through normal forms in free products that this paper does not use (Remark 5.6). The model also wrote the verification scripts `verify_general.py` and `make_figures.py`, and it drafted the proofs, the literature survey and the text, which the author then revised. I checked the proofs by prompting the same model to audit their logic line by line in separate passes. Every count and every rank in the paper was also checked by computer, to depth 28 for $m = 2$ and 30 for $m = 3$, by two independent methods (`verify_general.py`). All claims and proofs in this paper are the responsibility of the author.

References

[1] S. Fairchild and C. R. H. Hanusa, Densities of arithmetic Hecke triangle group orbits, preprint, 2026, https://arxiv.org/abs/2607.22142.

[2] E. Hecke, Über die Bestimmung Dirichletscher Reihen durch ihre Funktionalgleichung, *Math. Ann.* **112** (1936), 664–699.

[3] B.-H. Im and W. Lee, Non-holomorphic Eisenstein series for certain Fuchsian groups and class numbers, preprint, 2022, https://arxiv.org/abs/2207.05325.

[4] E. Janvresse, B. Rittaud, and T. de la Rue, Dynamics of λ-continued fractions and β-shifts, preprint, 2011, https://arxiv.org/abs/1103.6181.

[5] P. Kagey, Problem 137, Open Problem Collection, https://peterkagey.com/problems/137/.

[6] P. Kagey, Enumerating all fractions by $x \mapsto x + 1$ and $x \mapsto -\frac{1}{x}$, Mathematics Stack Exchange, question 5057812, April 2025, https://math.stackexchange.com/q/5057812.

[7] C. Kimberling and P. J. C. Moses, The infinite Fibonacci tree and other trees generated by rules, *Fibonacci Quart.* **52** (2014), no. 5, 136–150.

[8] A. Leutbecher, Über die Heckeschen Gruppen $G(\lambda)$. II, *Math. Ann.* **211** (1974), 63–86.

[9] mathmasterzach, Answer to "Enumerating all fractions by $x \mapsto x + 1$ and $x \mapsto -\frac{1}{x}$", Mathematics Stack Exchange, April 2025, https://math.stackexchange.com/a/5058406.

[10] OEIS Foundation Inc., Sequence A060961, The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A060961.

[11] C. Kimberling, Sequence A226247, The On-Line Encyclopedia of Integer Sequences, 2013, https://oeis.org/A226247.

[12] OEIS Foundation Inc., Sequence A293506, The On-Line Encyclopedia of Integer Sequences, https://oeis.org/A293506.

[13] L. Paris and O. Varghese, The growth series of Dyer groups, preprint, 2023, https://arxiv.org/abs/2305.09796.

[14] A. Pemmasani, The tree of rationals generated by $x \mapsto x+1$ and $x \mapsto -1/x$: an exact distance formula and the rank recurrence, preprint, 2026.

[15] D. Rosen, A class of continued fractions associated with certain properly discontinuous groups, *Duke Math. J.* **21** (1954), 549–563.

[16] D. Rosen and T. A. Schmidt, Hecke groups and continued fractions, *Bull. Austral. Math. Soc.* **46** (1992), 459–474.

[17] I. Short and M. Walker, Geodesic Rosen continued fractions, *Q. J. Math.* **67** (2016), 519–549.

[18] D. Taha, On cross sections to the geodesic and horocycle flows on quotients of $\mathrm{SL}(2,\mathbb{R})$ by Hecke triangle groups G_q, preprint, 2019, https://arxiv.org/abs/1906.07250.

[19] K. Winsor, Hecke triangle groups and special hyperbolic elements, preprint, 2026, https://arxiv.org/abs/2605.30064.